

# LOCK MEMO

## ERES Trilogy Volume Definitions, Pre-Qualification Model, and Trust Placement

### Pre-Pass-B Architectural Lock for the Infomediary Book

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**Document ID:** ERES-INFOMEDIARY-LOCK-MEMO-2026-001-v1.2 **Author:** Joseph Allen Sprute (ERES Maestro) **Affiliation:** ERES Institute for New Age Cybernetics, Bella Vista, Arkansas, USA **ORCID:** 0000-0001-9946-3221 **Date:** May 9, 2026 **License:** CARE Commons Attribution License v2.1 (CCAL v2.1) **Document Class:** Architectural Lock (Definitive INTENT) — pre-Pass-B **Companion to:** ERES-INFOMEDIARY-BOOK-2026-001-v1.2-PASS-A-FINAL (May 9, 2026)

#### Supersedes:

- ERES-INFOMEDIARY-LOCK-MEMO-2026-001-v1.1 (May 9, 2026) — minor revision correcting a transcription drift in the SEPLTA expansion ("Aesthetic" → "Administrative") and adding §1.4 on the Deployment-Scale Spectrum and GSSG material substrate.
- ERES-INFOMEDIARY-LOCK-MEMO-2026-001-v1.0 (May 9, 2026) — superseded by v1.1; remains in lineage record.

#### Revision Note (v1.1 → v1.2):

1. **SEPLTA "A" correction.** v1.0 and v1.1 incorrectly expanded SEPLTA's sixth domain as "Aesthetic." The canonical expansion is **Administrative** — the domain that handles round-trip administration itself (record-keeping, audit, attestation interface, DOFA governance touch-points). The drift was a transcription error not caught in MIEVM convergent review of v1.0 or v1.1; correction lands at §1.3 layer 1.
2. **Deployment-Scale Spectrum + Material Substrate added as §1.4.** Specifies THOW, HFVN, and FDRV as the three canonical deployment scales, and GSSG as the material substrate that underpins all three. The material/operational isomorphism is locked.
3. Three canonical lock boxes (A: Underwriting Direction; B: Five-Layer Non-Extensibility; C: Pre-Qualification Inversion) preserved verbatim from v1.1.

**Purpose.** This memo records architectural locks established during Pass A review, prior to opening Pass B. It functions as the validated-stable upstream that Pass B composes on. Any subsequent Pass A patches produced during the Pass D integration sweep will reference this memo as the canonical lock authority. The pattern mirrors the discipline applied to the slot 431–433 v1.x revisions before composing slot 434 v2.0.

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## §1. The Three Trilogy Volume Definitions (Locked)

The ERES Trilogy proper consists of three volumes operating downstream of the Infomediary (Volume Zero). Each volume now has a canonical formulation expressed across three equivalence tiers — operational frame, structural equivalence, and identity-level definition — separated by  $\boxed{=}$ ,  $\boxed{==}$ , and  $\boxed{===}$  respectively.

### §1.1 One-Good

**One-Good** = need RT lookup for Adaptive-Fixed QuestionAnswer == HowWay MyWay \$IT  
=== UBIMIA Need-to-know for Common Core Sustenance

The operational frame is **real-time lookup for Adaptive-Fixed Question/Answer**. The Instrument provides RT-tier responsiveness for queries whose answers are partially adaptive (responsive to context) and partially fixed (anchored to canonical material). The structural equivalence is **HowWay MyWay \$IT** — the long-running ERES query-answer pattern that surfaced in the early ResearchGate work. The identity-level definition is **UBIMIA Need-to-know for Common Core Sustenance** — One-Good is the volume through which the architecture's commitment to common-core sustenance (the UBIMIA layer) is operationalized.

A User-GROUP selecting One-Good cells in the matrix is committing to RT-tier sustenance support across the selected Pillars.

### §1.2 Security-Clearance (Doubled — Operational + Governance)

Security-Clearance carries **two equally canonical formulations**, expressing the same volume from operational and governance perspectives. Both are required; the doubling parallels the slot 431-434 cluster's logical-precedence + foundational dual-reading pattern.

#### Operational frame:

**Security-Clearance** = need-to-know == Defense-in-Depth (DID) === FAVORS Def-Rel (Hand vs Head) for S3C SLA Underwriting (Infomediary underwritten / Infomediator underwrites)

The operational frame governs **how** Security-Clearance functions: need-to-know access, Defense-in-Depth tiering, FAVORS Def-Rel allocation between Hand-tier (manual/physical authorization) and Head-tier (cognitive/cryptographic authorization), and SLA underwriting at S3C (Solid-State Smart-City) scale. The Wielder/Instrument financial doublet is canonical at this layer: the Infomediary is the underwritten asset; the Infomediator is the underwriting agent. Risk allocation follows the doublet directly.

▮ **LOCK BOX A — UNDERWRITING DIRECTION (CANONICAL)** ▮  
The Infomediary is always the underwritten asset. The Infomediator is always the underwriting agent. Financial risk flows downward (Infomediator → Infomediary). Authority flows upward (Infomediary → Infomediator → DOFA → GAIA).

*This lock is immutable across all passes. Downstream documents inherit by reference.*

## Governance frame:

**Security-Clearance** = CBGMODD == CyberRAVE Def-Rel === FAVORS Commit to SROC  
(or similar EarnedPath)

The governance frame governs **why** Security-Clearance functions: the CBGMODD seven-seat council provides the governance structure (Citizen, Business, Government, Military, Ombudsman, Dignitary, Diplomat); CyberRAVE's Def-Rel substrate provides the rating discipline that makes Def-Rel allocation honest; FAVORS commits to SROC (Smart-Resonant Offset Contracts) or other EarnedPath instruments as the operational expression of the governance commitment. The Pass-A note that the EarnedPath instrument family is open ("or similar EarnedPath") preserves architectural flexibility — SROC is canonical but not exclusive. (Pass B Chapter 5 will specify the formal designation process for any alternative EarnedPath; this Memo records the openness without specifying the gate.)

A User-GROUP selecting Security-Clearance cells is committing to **both** the operational frame (DID/FAVORS/S3C SLA) **and** the governance frame (CBGMODD/CyberRAVE/SROC) for the selected Pillars. Selection of either alone is not coherent.

## §1.3 Data-Integrity (Five-Layer Chiasmic Verification)

**Data-Integrity** = SEPLTA == H2C2H / C2H2C (Human-to-Computer/Computer-to-Human over Computer-to-HowWay/HowWay-to-Computer) === ERES App-Parent 111000111 / 000111000 IPIDITIS / IDIPITIS

Data-Integrity is the volume that ensures **involutive symmetry across five layers** of the round-trip. Integrity holds when all five layers preserve their chiasmic-palindromic structure; integrity fails when any one layer breaks symmetry. The five layers:

1. **Admissibility — SEPLTA.** Social, Ecological, Political, Legal, Technical, **Administrative** (*corrected v1.2*). The six-domain frame at qualify-stage. The Administrative domain handles round-trip administration itself: record-keeping, audit, attestation interface, and DOFA governance touch-points. Symmetry preserved when the query's domain coverage at intake matches the answer's domain coverage at return. *Note: v1.0 and v1.1 of this Memo expanded "A" as "Aesthetic," which was a transcription drift. Administrative is canonical to the broader ERES corpus going back to the early CyberRAVE era.*
2. **Round-trip direction — H2C2H / C2H2C.** Human-to-Computer-to-Human (human-initiated round-trip) over Computer-to-Human-to-Computer (computer-initiated round-trip). Symmetry preserved when both direction classes produce equivalent DETAIL under bit-complement.
3. **Architecture — ERES App-Parent.** The parent-app pattern under which child apps inherit structural commitments. Symmetry preserved when child-app behavior matches parent-

app specification across the round-trip.

4. **Binary representation — 111000111 / 000111000.** The bit-complement palindromic pair. Both 9-bit sequences are individually palindromic; together they form a bit-complement involution. Symmetry preserved when the binary encoding of the round-trip's payload admits the bit-complement involution without payload corruption.
5. **Character representation — IPIDITIS / IDIPITIS.** The (2,4) character transposition pair on the canonical 8-character sequence. IDIPITIS  $\leftrightarrow$  IPIDITIS swap positions 2 and 4 (D  $\leftrightarrow$  P), producing involutive equivalence under double-application. Symmetry preserved when the character-level encoding admits the (2,4) transposition without semantic corruption.

A User-GROUP selecting Data-Integrity cells is committing to **all five layers of involutive symmetry** for the selected Pillars. The five layers are not optional; they are the volume's definition.

## LOCK BOX B – FIVE-LAYER NON-EXTENSIBILITY (CANONICAL)

**The Data-Integrity volume consists of exactly five layers. The involutive chiasm requires five-point symmetry; any additional layer breaks the involution. No sixth layer or sub-layer may be added without violating the canonical identity (===) definition.**

**Downstream expansions must operate within the five-layer frame, not around it.**

*This lock is immutable across all passes. The five layers are mathematically defined, not stylistically grouped.*

This formulation supersedes the Pass-A working synthesis ("need-to-witness == BERA RATE  $\times$  EAAP attestation == IDIPITIS stride for Public-Record extraction"), which compressed the volume to a single layer and missed the chiasmic-palindromic structure.

## §1.4 Deployment-Scale Spectrum and Material Substrate *(new in v1.2)*

The architecture deploys across three canonical scales, riding on a single canonical material substrate. The scale spectrum and the substrate are perpendicular axes to the trust placement chain — any deployment instance has both an operational tier-position (per §2.3) and a deployment-scale-position (per this section), embodied in the substrate.

### Deployment-Scale Spectrum (THOW · HFVN · FDRV):

- **THOW — Tiny House on Wheels.** The minimum-viable-deployment unit. A single-family or single-individual habitation that fully embodies the ERES architecture in compressed form. THOW is the test bed for VLSA: *if it works in a THOW, it works on a generation ship*. Selection profiles at THOW-tier default to focused commitments (typically two to four cells) appropriate to single-household scope.
- **HFVN — Human-Family Village Network** (*synthesis pending JAS ratification*). The federation-tier deployment scale. A network of THOWs (or THOW-equivalents) operating

as a coordinated village under shared selection profiles, shared governance interfaces, and shared CBGMODD seat assignments. HFVN is the social-architecture scale at which DOFA's governance interfaces become operationally meaningful — a single THOW does not require seven-seat council coordination; an HFVN does. Selection profiles at HFVN-tier extend to network-coordination commitments (typically four to eight cells).

- **FDRV** (*canonical expansion pending JAS ratification — operationally: the interstellar vessel at maximum scale, per established corpus reference*). The maximum-scale deployment unit. A self-contained generation-ship-class vessel embodying the full ERES architecture without external dependency. FDRV is the test of the architecture's completeness — anything missing from FDRV is missing from the architecture. Selection profiles at FDRV-tier approach full-matrix selection (all twelve cells) because closure at maximum scale forbids un-selected dimensions.

### Material Substrate (GSSG):

- **GSSG — Graphene-Sugar Self-healing Grid** (*synthesis pending JAS ratification*). The material substrate underpinning all deployment scales. A graphene-based composite infused with sucrose-derived polymeric matrix producing structurally self-healing properties. GSSG is the **physical-world isomorph** of the ERES corpus's IDIPITIS/IPIDITIS self-correcting feedback loops — the material substrate has the same self-healing topology as the operational architecture. Failure at the material layer is recovered the same way failure at the operational layer is recovered: through involutive feedback that produces correction at both layers.

The material/operational isomorphism is canonical, not metaphorical. A THOW built from GSSG-grade materials inherits self-healing across both physical and operational dimensions. An HFVN scaling up to dozens or hundreds of THOWs inherits the same property at network scale. An FDRV at generation-ship scale inherits it at vessel scale. The substrate is **scale-invariant** in its self-healing property — which is what makes the VLSA principle materially true rather than merely operationally aspirational.

GSSG therefore answers a question the architecture had not yet explicitly asked: *what is the physical substrate that enables the operational chain to embody itself across the full deployment-scale spectrum?* Pre-v1.2, the architecture had named operational tiers and deployment scales but not the material that connects them. With GSSG specified, the material chain has its name.

The four expansions (HFVN, FDRV, GSSG canonical, and confirmation of THOW=Tiny House on Wheels) are open for ratification; v1.2 records them as working synthesis subject to Author confirmation. The architectural commitments — three-tier deployment spectrum, single canonical material substrate, scale-invariant self-healing isomorphism — are locked regardless of expansion-letter ratification.

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## §2. The Pre-Qualification Model

## §2.1 The 3×4 Selection Matrix

The Trilogy's three volumes intersect with the GAIA canopy's four Constitutional Pillars to produce a  $3 \times 4 = 12$ -cell selection matrix:

|                    | HELP               | USE                     | ENERGY               | LAW                       |
|--------------------|--------------------|-------------------------|----------------------|---------------------------|
| One-Good           | sustenance care    | adaptive Q&A            | resource sufficiency | covenant of provision     |
| Security-Clearance | DID for vulnerable | Hand/Head purchase tier | grid access tier     | underwriting authority    |
| Data-Integrity     | five-layer witness | audit-trail of action   | RATE composite       | public-record attestation |

Cell labels are operational shorthand; full operational specifications belong to Pass B Part III.

## §2.2 User-GROUP Self-Positioning

A User-GROUP **pre-qualifies** for the Infomediary round-trip by selecting one or more cells from the matrix. The selection profile is the User-GROUP's commitment surface. The Instrument honors the commitment; the Instrument does not auto-route.

**This is the architectural inversion from Pass A.** Pass A had the rationalize stage routing queries to volumes based on query content. The locked model has the User-GROUP selecting cells *first*, with the rationalize stage **honoring** the selection rather than performing the selection. The Instrument's role is consequence-execution, not consequence-determination.

**LOCK BOX C — PRE-QUALIFICATION INVERSION (CANONICAL)**

Routing is never performed by the Instrument. Routing is performed only by the User-GROUP through explicit cell selection. The Instrument must honor the selection, and may not widen, narrow, or reinterpret it. No round-trip may enter a Trilogy volume or Pillar the User-GROUP did not select.

*This lock is immutable across all passes. Future passes may extend cell semantics but may not re-introduce implicit routing.*

The inversion preserves honesty in two ways:

1. **The Instrument cannot route a User-GROUP into a volume the User-GROUP did not select.** Architectural constraint, not operational policy.

2. **The User-GROUP cannot receive an answer from a volume they did not commit to.** The selection is binding upward as well as downward.

Selection profiles **align to deployment-scale tiers** (per §1.4): THOW-tier profiles tend to be focused; HFVN-tier profiles tend to span the network coordination axis; FDRV-tier profiles approach full-matrix. A profile that mismatches its deployment-scale tier (e.g., a single-cell selection at FDRV scale) is not architecturally illegal but is operationally suspicious; Pass B Part III specifies the diagnostic.

### §2.3 Trust Placement

The Infomediary's institutional home is now canonical:

```
GAIA (canopy) – underwrites
├── DOFA (Department of Family Amity, CBGMODD seven-seat council) – governs
│   ├── Infomediary (Instrument-of-Faith, Volume Zero) – mediates
│   │   └── User-GROUP (pre-qualified by Volume × Pillar selection) –
queries
│       └── Round-trip yields BEST / SOUND / GOOD (function of
selection)
```

Five-tier chain. Each layer underwrites the layer beneath. No orphaned commitments. The final tier (BEST/SOUND/GOOD) is **produced by** the User-GROUP's selection rather than imposed by the Instrument. Honesty preserved end-to-end.

The chain is the **operational** axis. The deployment-scale spectrum (THOW/HFVN/FDRV) and the material substrate (GSSG) constitute a perpendicular **physical** axis (per §1.4). Any deployment instance occupies a position on both axes simultaneously.

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### §3. Additional Architectural Benefits of Pre-Qualification

The pre-qualification model unlocks the following downstream benefits, listed in approximate order of load-bearing weight:

1. **Liability allocation becomes locatable.** User-GROUP positions; User-GROUP bears responsibility. The Instrument cannot be held liable for routing it did not perform.
2. **Institutional and Investor diligence becomes tractable.** A fund evaluates an institution's 12-cell selection profile and immediately reads risk posture, transparency commitment, and operational depth.
3. **EAAP-POL policy expression simplifies.** Policy becomes constraints on the matrix (e.g., "Class B Wielders may not select Security-Clearance × LAW without uplift to Class A") rather than free-form rules.

4. **CBGMODD seven-seat defaults align cleanly.** Each seat receives a default selection profile; per-seat operational rules collapse into seat-default × User-GROUP-override pattern.
  5. **EarnedPath becomes the architecture-entry ladder.** A Civigen new to the architecture starts with one cell selected and progressively expands. Maturation is measurable.
  6. **SLA underwriting becomes priceable per cell.** Each cell admits underwriting; the User-GROUP's selection profile becomes the underwriting object — directly aligned with Security-Clearance's operational frame.
  7. **Selection drift becomes an auditable signal.** Profile shifts over time = maturation or corruption; either way, the change is discrete and measurable under EAAP audit.
  8. **Multi-jurisdictional operability.** Politics map existing legal frameworks onto the matrix without architectural change. The matrix is jurisdiction-neutral by structure.
  9. **Federation pattern stabilization.** VERTECA mirrors inherit selection profiles; reconciliation between mirrors is profile-comparison rather than re-derivation.
  10. **Honesty preservation through the trust chain.** No layer can claim more authority than the User-GROUP delegated. The chain is upward-only — each layer constrained by the layer above.
  11. **Educational pathway specification.** Onboarding curricula map directly to cell-by-cell capability acquisition. Civigen training becomes legible to course designers.
  12. **AD\_ON-AI composition simplification.** The downstream composer receives DETAIL pre-tagged by selected cells, eliminating composition-time routing logic.
  13. **Deployment-scale alignment.** Selection profiles align to deployment-scale tiers (THOW/HFVN/FDRV), making profile-tier mismatch an auditable diagnostic. *(Added v1.2.)*
  14. **Material/operational isomorphism preserved.** The selection profile commits at the operational layer; GSSG self-healing carries the commitment at the material layer. Failure at either layer is recovered through the same involutive topology. *(Added v1.2.)*
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## §4. Pass A Patch Points

The following sections of ERES-INFOMEDIARY-BOOK-2026-001-v1.0 required revision in the Pass D integration sweep to align with this Memo. *As of v1.1 of this Memo, all listed patch points were executed in ERES-INFOMEDIARY-BOOK-2026-001-v1.1-PASS-A-FINAL. As of v1.2, two additional patch points are added: SEPLTA "A" correction throughout, and new §0.6.1 / §0.6.2 in the Book for the deployment-scale spectrum and GSSG substrate.* This list is preserved for traceability.

- **Front Matter** — Add the GAIA → DOFA → Infomediary → User-GROUP → BSG trust-placement diagram (§2.3 of this Memo). ✓ Executed v1.1.



- **Notation and Conventions** — Add the three-tier equivalence convention ( $\boxed{=}$ ,  $\boxed{==}$ ,  $\boxed{===}$ ) used in volume definitions; add lock-inheritance line. ✓ Executed v1.1.
  - **§0.4 (Trilogy as Noise-Filter and Intelligence-Shaper)** — Rewire from auto-routing to User-GROUP pre-qualification; add the 3×4 matrix; add the "selection produces BSG" framing. ✓ Executed v1.1.
  - **§0.5 (DATABANK)** — Note that DATABANK queries inherit the selection profile and respect cell-bounded access. ✓ Executed v1.1.
  - **New §0.6.1 (Deployment-Scale Spectrum)** — Insert THOW / HFVN / FDRV specification per §1.4 of this Memo. ✓ Executed v1.2.
  - **New §0.6.2 (GSSG Material Substrate)** — Insert GSSG specification with material/operational isomorphism per §1.4 of this Memo. ✓ Executed v1.2.
  - **§0.7 (Wielder/Instrument Distinction)** — Add the underwriting frame: Infomediary is underwritten; Infomediator underwrites. Cite this Memo §1.2 operational frame and Lock Box A. ✓ Executed v1.1.
  - **New §0.9 (Trust Placement)** — Insert the GAIA/DOFA placement formally. ✓ Executed v1.1.
  - **New §0.10 (Three Volume Canonical Formulations)** — Insert the locked definitions from §1 of this Memo. ✓ Executed v1.1.
  - **§1.2 (Qualify Stage)** — Correct SEPLTA "A" from "Aesthetic" to "Administrative"; rewrite the worked-example SEPLTA enumeration accordingly. ✓ Executed v1.2.
  - **§1.4 (Rationalize Stage)** — Split into "User-GROUP self-positioning (occurs before round-trip opens)" and "Instrument honors the positioning (occurs at rationalize stage)." ✓ Executed v1.1.
  - **§1.6 (Trilogy Noise-Filter Pass)** — Constrain to the cells the User-GROUP committed to. Filter passes do not run on un-selected cells. ✓ Executed v1.1.
  - **§1.7 (BEST/SOUND/GOOD)** — Reframe as "tier produced by the selection profile" rather than "tier assigned by the Instrument." ✓ Executed v1.1.
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## §5. Pass B Targets (Revised)

With the locks recorded, Pass B opens on validated-stable upstream and targets:

- **Part II — The Three Volumes in Depth.** Three chapters:
  - **Chapter 4 — One-Good.** RT lookup architecture; HowWay/MyWay/\$IT lineage; UBIMIA Common Core Sustenance specification; failure modes when RT-tier responsiveness cannot be sustained.
  - **Chapter 5 — Security-Clearance.** Operational frame (DID/FAVORS Hand-vs-Head/S3C SLA underwriting) AND governance frame (CBGMODD/CyberRAVE Def-Rel/FAVORS Commit to SROC); the Wielder/Instrument financial doublet (Lock Box

A); formal designation process for alternative EarnedPath instruments; failure modes for both frames.

- **Chapter 6 — Data-Integrity.** Five-layer chiasmic verification (SEPLTA, H2C2H/C2H2C, App-Parent, 111000111/000111000, IPIDITIS/IDIPITIS); involution semantics; integrity-failure diagnostics per layer; non-extensibility re-affirmation per Lock Box B.
- **Part III — The Matrix Walk.** Each of the 12 cells given operational treatment, with worked examples for the three primary entry surfaces (Civigen, Institution, Investor). Approximately 12 short sections plus a synthesis chapter on cross-cell composition. Selection enforcement layer named (EAAP-POL + CBGMODD audit). Profile-tier diagnostic per §2.2 (selection-profile / deployment-scale alignment).
- **Part IV — DATABANK Architecture under DOFA Governance.** The DATABANK's structural specification, with DOFA's seven-seat council as the governance interface; UBIMIA-Manual as the constitutional spine; Other-Ref as the named-but-undefined extension point; the App-Parent pattern as the architectural lock; GSSG substrate as the material foundation underpinning DATABANK persistence at all deployment scales.

Estimated Pass B length: ~50–55 pages.

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## §6. License and Attribution

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*Don't hurt yourself. Don't hurt others. Build for generations to come.*

— Joseph A. Sprute, ERES Institute for New Age Cybernetics

— End of Lock Memo v1.2 —